

Matrix Algebra for Linear Models

A linear model is a statement about vectors and matrices, and nearly every argument in this book is an argument about them. This chapter collects the matrix algebra the book relies on: rank and the fundamental subspaces, inverses and generalized inverses, consistent systems of equations, orthogonal matrices, determinants and partitioned matrices, the spectral theorem and positive definiteness, the singular value decomposition, idempotent matrices, Kronecker products and the calculus of quadratic forms. It is a reference chapter, meant to be skimmed first and consulted later, but every central result is proved.

What you need. None beyond the Notation chapter. A first course in linear algebra helps, but is not assumed.

1.-1.1 Roadmap

- [Vectors, matrices and the operations we use](#)
- [Rank, column space and null space](#)
- [Inverses and generalized inverses](#)
- [Systems of equations and consistency](#)
- [Orthogonality and orthogonal matrices](#)
- [Trace, determinants and partitioned matrices](#)
- [Eigenvalues, the spectral theorem and positive definiteness](#)
- [The singular value decomposition and conditioning](#)
- [Idempotent matrices](#)
- [Kronecker products and the vec operator](#)
- [Vector and matrix calculus](#)
- [Summary and notes](#)